



Lab 09

Passive Filters

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9.1 Objective

To design and build passive Low Pass, High Pass and Band Pass Filters and verify with an appropriate implementation.

9.2 Introduction

A filter is a device that changes the amplitude (height) of an AC voltage (a voltage in the form of a sine wave) as the frequency of the input voltage changes. Filters have two terminals. The input terminals take in the input voltage, which passes through the filter and onto the output terminals, where the resulting output waveform can be observed.

A filter is a circuit capable of passing (or amplifying) certain frequencies while attenuating other frequencies. Thus, a filter can extract important frequencies from signals that also contain undesirable or irrelevant frequencies. Filters can be placed in one of two categories: passive or active.

Passive filters include only passive components resistors, capacitors, and inductors. In contrast, active filters use active components, such as op-amps, in addition to resistors and capacitors, but not inductors. Passive filters are most responsive to a frequency range from roughly 100 Hz to 300 MHz. Active filters are capable of dealing with very low frequencies (approaching 0 Hz), and they can provide voltage gain (passive filters cannot).

The four primary types of filters include the low-pass filter, the high-pass filter, the band-pass filter, and the notch filter (or the band-reject or band-stop filter). Take note, however, that the terms “low” and “high” do not refer to any absolute values of frequency, but rather they are relative values with respect to the cutoff frequency Figure 9.1 below gives a general idea of how different types of filters works:

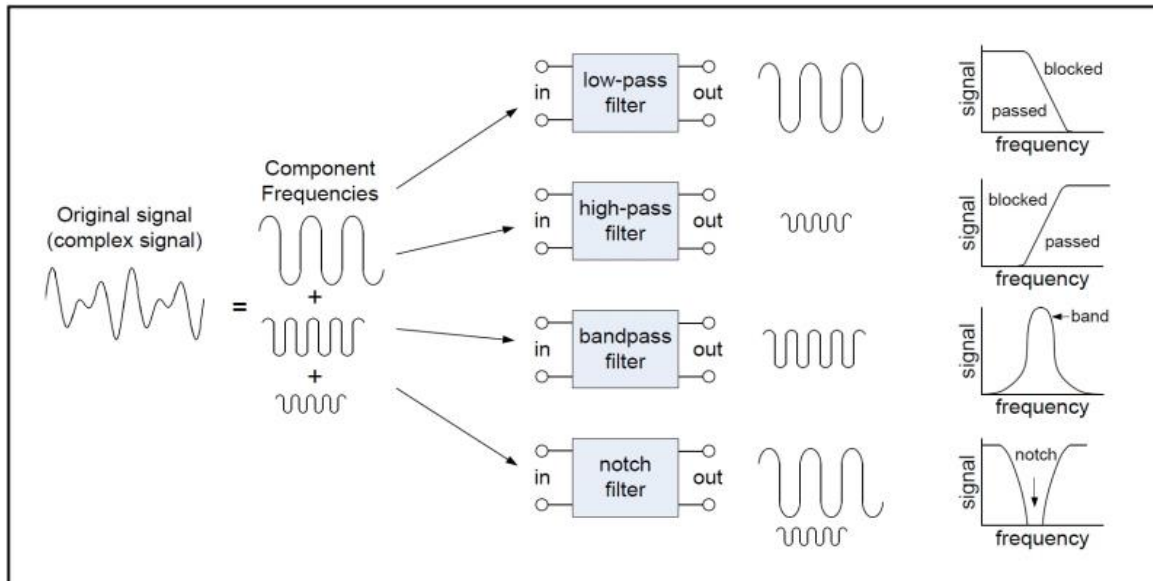


Figure 9.1: Types of filters

9.3 Low Pass Filter

Low pass filter is an electronic circuit which allows the **low frequency to pass** through it and attenuates the high-frequency signal. The low pass filter offers low reactance to the signals with frequencies lower than the cut-off frequency so that low frequencies can pass but it provides high reactance to the high-frequency signal and thus block them. This type of filter is known generally as a “first-order filter” or “one-pole filter” because it has only “one” reactive component, the capacitor, in the circuit.

9.3.1 Cut-off Frequency

Cut-off, Corner or Breakpoint frequency is defined as being the frequency point where the capacitive reactance and resistance are equal, $R = X_c$. When this occurs the output signal is attenuated to 70.7% of the input signal value or $-3dB$ ($20 \log (V_{out}/V_{in})$) of the input. The significance of cut-off frequency is that they define the useful or important part of the spectrum. Beyond the cutoff frequency, the filter will attenuate all other frequencies, depending on the design of the filter. The gain of circuit at cut-off frequency is always equal or approximate to the value of $\sqrt{1/2}$ or 0.707. As the filter contains a capacitor, the Phase Angle (Φ) of the output signal lags behind that of the input and at the -3dB cut-off frequency (f_c) is -45° out of phase.

The reactance of a capacitor varies inversely with frequency, while the value of the resistor remains constant as the frequency changes. At low frequencies the capacitive reactance, (X_c) of the capacitor will be very large compared to the resistive value of the resistor, R . This means that the voltage potential V_c across the capacitor will be much larger than the voltage drop V_R developed across the resistor. The cut off frequency for Low pass RC circuits is given as:

$$f_c = \frac{1}{2\pi RC}$$

9.3.2 Frequency Response of Low Pass Filter

The Bode Plot shows the Frequency Response of the filter to be nearly flat for low frequencies and all of the input signal is passed directly to the output, resulting in a gain of nearly 1, called unity, until it reaches its Cut-off Frequency point (f_c). This is because the reactance of the capacitor is high at low frequencies and blocks any current flow through the capacitor.

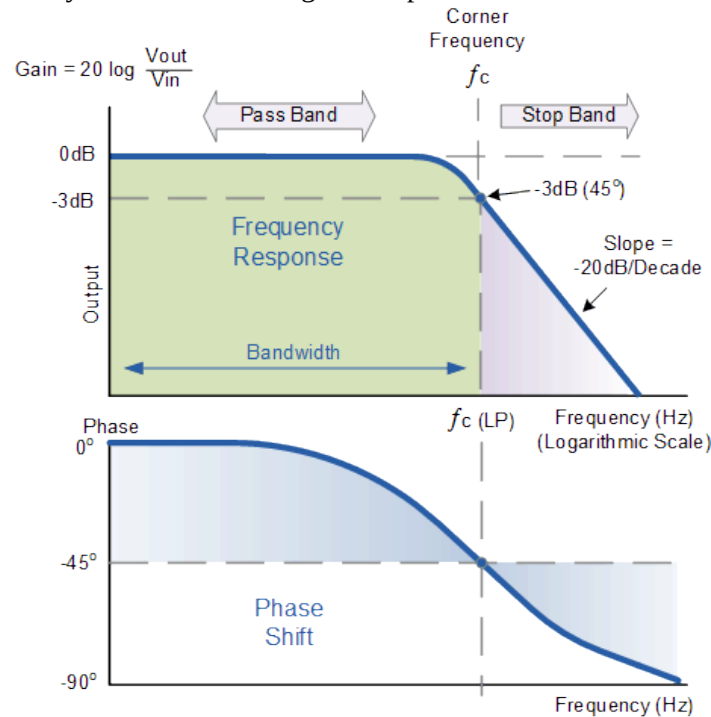


Figure 9.2: Frequency Response of Low Pass Filter

Task 1: To build passive Low Pass Filter and verify with hardware implementation

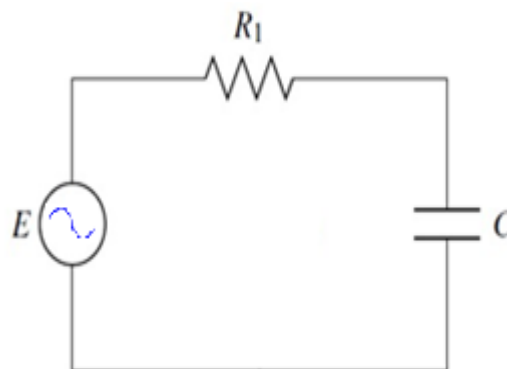


Figure 9.3: Circuit Diagram of Passive Low Pass Filter

1. Build the passive low-pass filter circuit shown in Figure 9.3 using the breadboard and the specified components. ($R = 10K\Omega$, and $C = 10nF$.)



- Set the function generator to output a 100 Hz sine wave with an amplitude of $3.5 V_{pk} - pk$. Connect the output of the function generator to the input of the circuit (V_{in}) and the oscilloscope probe to the output of the circuit (V_{out}).
- Adjust the oscilloscope settings to display the input and output waveforms on the same screen.
- Measure the amplitude and phase shift of the input and output waveforms.
- Repeat the same steps for the frequencies mentioned in Table 9.1.
- Measure the actual resistance and capacitance of the used components.

$$R = \underline{\hspace{2cm}} \quad 9.9k$$

$$C = \underline{\hspace{2cm}} \quad 10.4 \text{ nF}$$

- Calculate the cutoff frequency of the filter using the formula $f_c = 1 / (2\pi RC)$.
 $f_c = \underline{\hspace{2cm}} 1591.54 \text{ Hz}$
- Verify the cutoff frequency by measuring the amplitude and phase shift of the output waveform at frequencies both below and above the calculated cutoff frequency.
- Draw Magnitude (dB) vs Frequency and Phase vs Frequency Plots with proper labelling in the graphs given below:

Note: The function generator and oscilloscope settings may need to be adjusted to obtain accurate measurements, depending on the characteristics of the components used in the circuit. It is important to take accurate measurements and record your observations carefully in order to properly understand the behavior of the filters.

Table 9.1: Data Table for Low Pass Filter

Frequency (Hz)	100	200	500	800	1k	1.2k	1.5k	3k	10k	20k	30k
Vin Max (V)	3.6	3.56	3.52	3.52	3.52	3.52	3.52	3.52	3.52	3.52	3.52
Vout Max (V)	3.4	3.48	3.28	2.96	2.76	2.56	2.28	1.40	472 m	240 m	160m
ΔT	- 0.11 m	- 0.13 m	- 148u	-96u	-94u	-93u	-91u	-59u	- 22.6 u	- 11.1 u	-8u
Φ	-3.96	- 8.136	- 26.6 4	- 27.6 5	- 33.8 4	- 40.1 76	- 49.1 4	- 63.7 2	- 81.3 6	- 79.9 2	-86.4
$T = \frac{V_{out} \text{ Max}}{V_{in} \text{ Max}}$	0.944	0.978	0.93 2	0.84 1	0.78 4	0.72 7	0.64 8	0.39 8	0.13 4	0.06 82	0.045 5
$g(f) = 20\log(T)$	- 0.496	- 0.193	- 0.61 2	-1.5	-2.11	- 2.77	- 3.77	-8	- 17.4 6	- 23.3 2	- 26.84

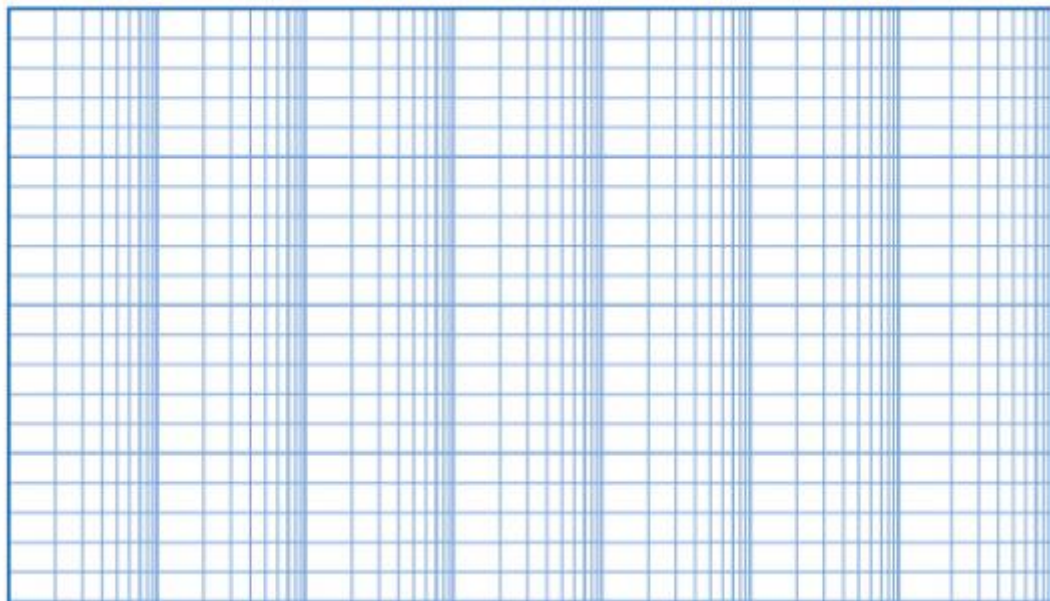


Figure 9.4: Gain(dB) Vs Frequency Graph for Low Pass Filter

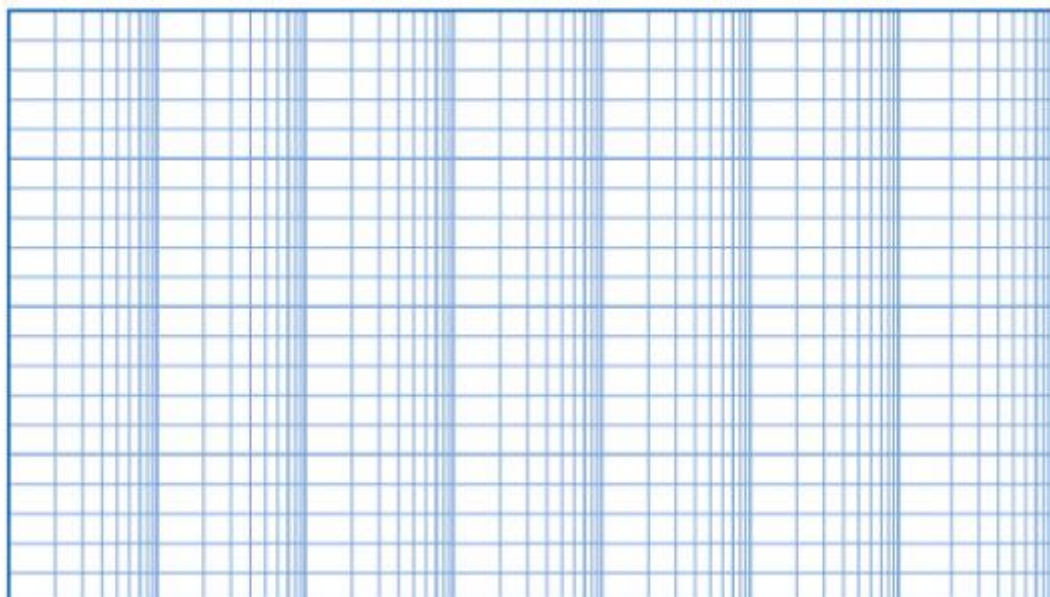
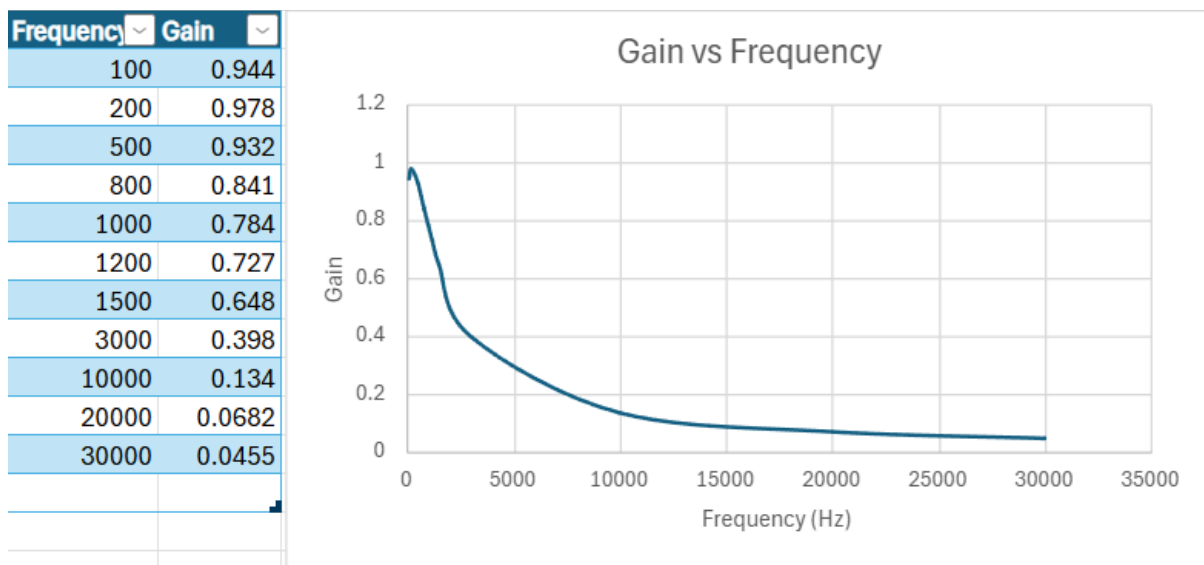
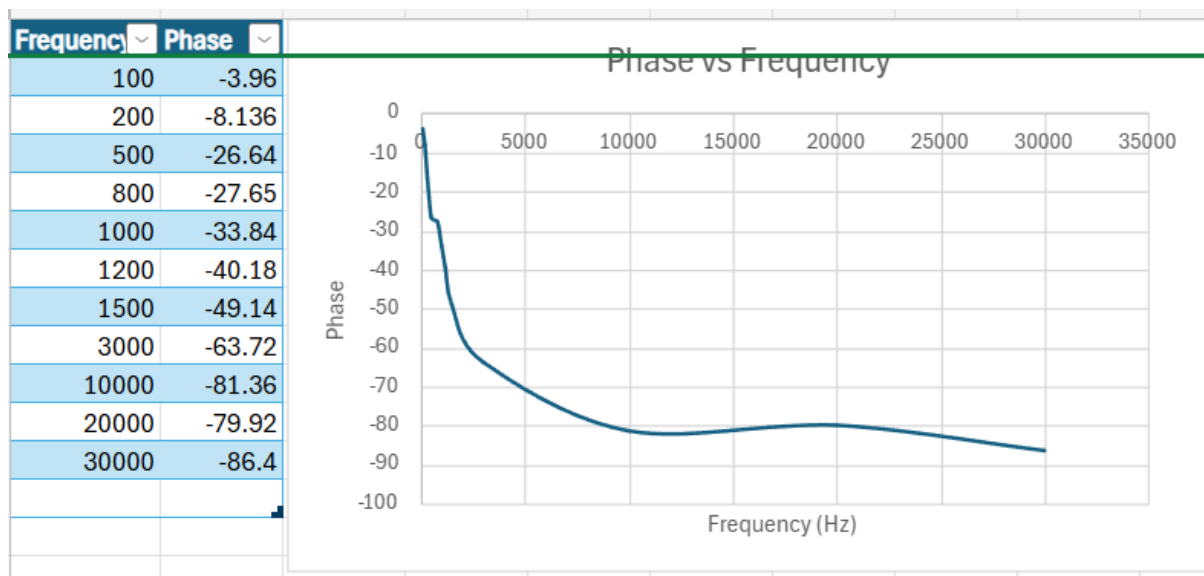


Figure 9.5: Phase Vs Frequency Graph for Low Pass Filter

Gain vs Frequency:



Phase vs Frequency:



9.4. High Pass Filter

A high pass filter is a circuit which allows the high frequency to pass through it while it will block low frequencies. Although it offers attenuation to high-frequency signal too but the attenuation factor is so small that it can be neglected. The high pass circuit consists of capacitor followed by a resistor. The input signals are applied to the capacitor, and then the voltage across the resistor is obtained as the output voltage.

The reactance of the capacitor is very high at low frequencies so the capacitor acts like an open circuit and blocks any input signals at V_{IN} until the cut-off frequency point (f_c) is reached. Above this cut-off frequency point the reactance of the capacitor has reduced sufficiently as to now act more like a short circuit allowing all of the input signal to pass directly to the output. High pass circuit is significant when

the distortion due to low frequency signal such as noise is to be removed. The cut off frequency for high pass RC circuits is given as:

$$f_c = \frac{1}{2\pi RC}$$

9.4.2 Frequency Response of High Pass Filter

The Bode Plot shows the Frequency Response of the filter to be nearly flat for high frequencies and all of the input signal is passed directly to the output, resulting in a gain of nearly 1, called unity. However, below the Cut-off Frequency point (f_c), there is attenuation.

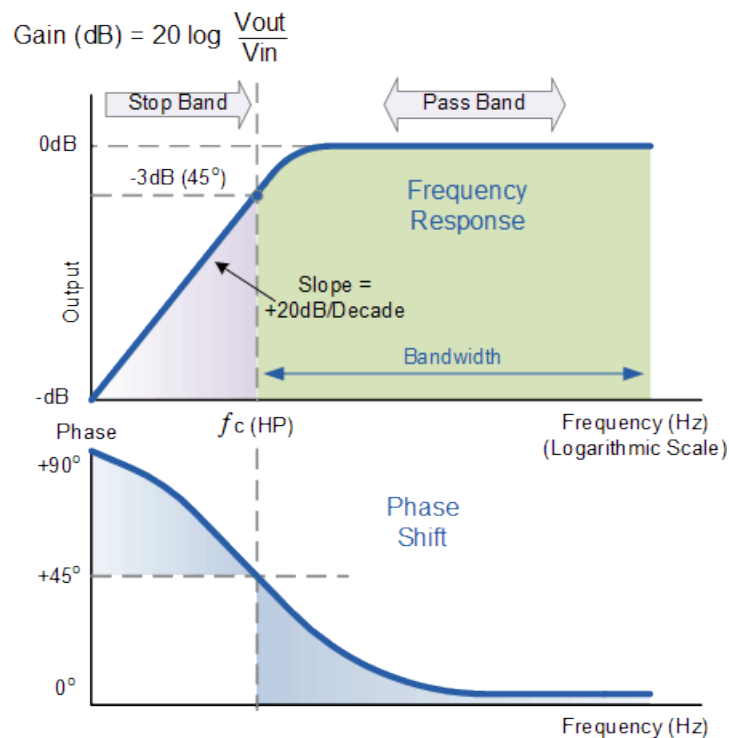


Figure 9.6 : Frequency Response of High Pass filter

Task 2: To build passive High Pass Filter and verify with hardware implementation

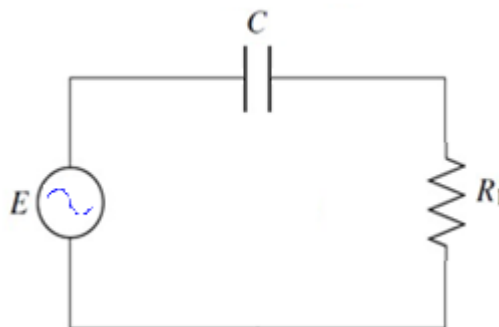


Figure 9.7: Circuit Diagram of Passive High Pass Filter



1. Build the passive high-pass filter circuit shown in Figure 9.7, using the breadboard and the specified components. ($R = 10K\Omega$, and $C = 10nF$.)
2. Set the function generator to output a 300 Hz sine wave with an amplitude of 3.5 V_{pk} – pk. Connect the output of the function generator to the input of the circuit (V_{in}) and the oscilloscope probe to the output of the circuit (V_{out}).
3. Adjust the oscilloscope settings to display the input and output waveforms on the same screen.
4. Measure the amplitude and phase shift of the input and output waveforms.
5. Repeat the same steps for the frequencies mentioned in Table 9.2.
6. Measure the actual resistance and capacitance of the used components.

$$R = \underline{\hspace{2cm}} 9.9k$$

$$C = \underline{\hspace{2cm}} 10.43 nF$$

7. Calculate the cutoff frequency of the filter using the formula $f_c = 1 / (2\pi RC)$.
 $f_c = \underline{\hspace{2cm}} 1591.4 \text{ Hz}$
8. Verify the cutoff frequency by measuring the amplitude and phase shift of the output waveform at frequencies both below and above the calculated cutoff frequency.
9. Draw Magnitude (dB) vs Frequency and Phase vs Frequency Plots with proper labelling in the graphs given below:

Table 9.2: Data Table for High Pass Filter

Frequency (Hz)	300	500	800	1k	1.2k	1.5k	2k	2.5k	5k	10k	50k
V_{in} Max (V)	3.52	3.52	3.52	3.52	3.52	3.52	3.52	3.52	3.52	3.52	3.52
V_{out} Max (V)	840 m	1.26	1.82	2.1	2.34	2.6	2.86	3.08	3.34	3.44	3.42
ΔT	780 u	388 u	206 u	148 u	110 u	77u	45u	29.5 u	7.4u	1.4u	72n
Φ	84.2 4	69.8 4	59.3 3	53.2 8	47.5 2	41.5 8	32.4	26.5 5	13.3 2	5.04	1.29 6
$T = \frac{V_{out} \text{ Max}}{V_{in} \text{ Max}}$	0.23 9	0.35 8	0.51 7	0.59 7	0.66 5	0.73 9	0.81 25	0.87 5	0.94 9	0.97 7	0.97 2
$g(f) = 20\log(T)$	- 12.4 6	- 0.89 2	- 0.57 3	- 0.44 8	- 0.35 4	- 0.26 3	- 0.18	- 0.11 6	- 0.04 55	- 0.02	- 0.02 5

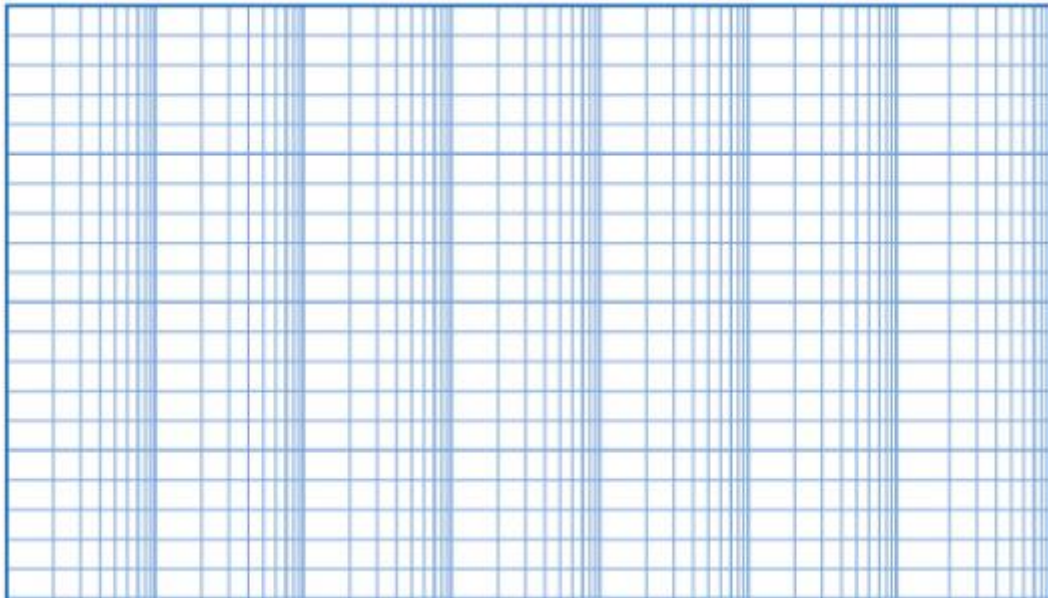


Figure 9.8: Gain(dB) Vs Frequency Graph for High Pass Filter

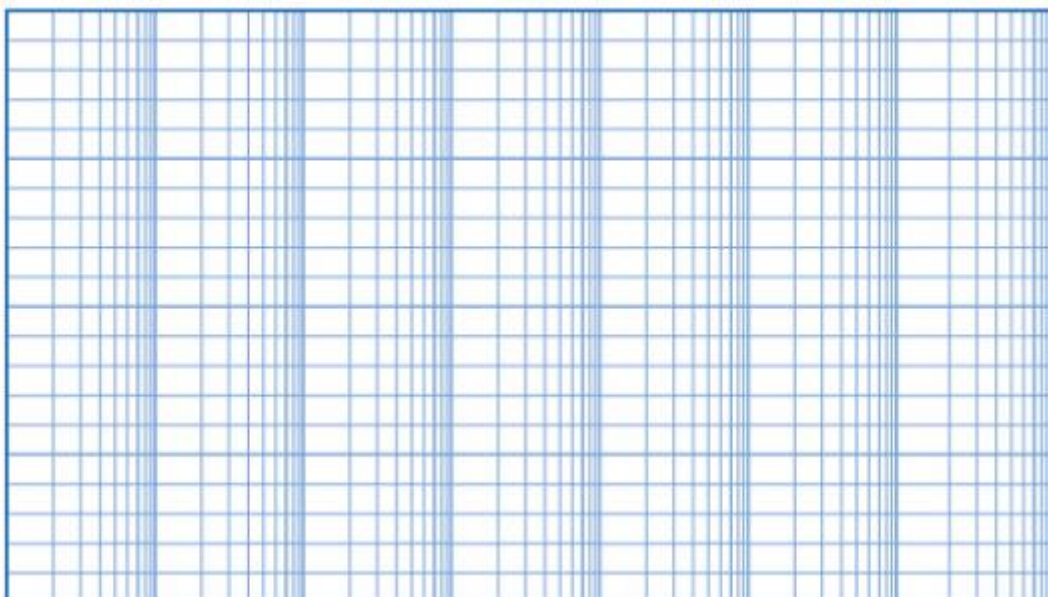
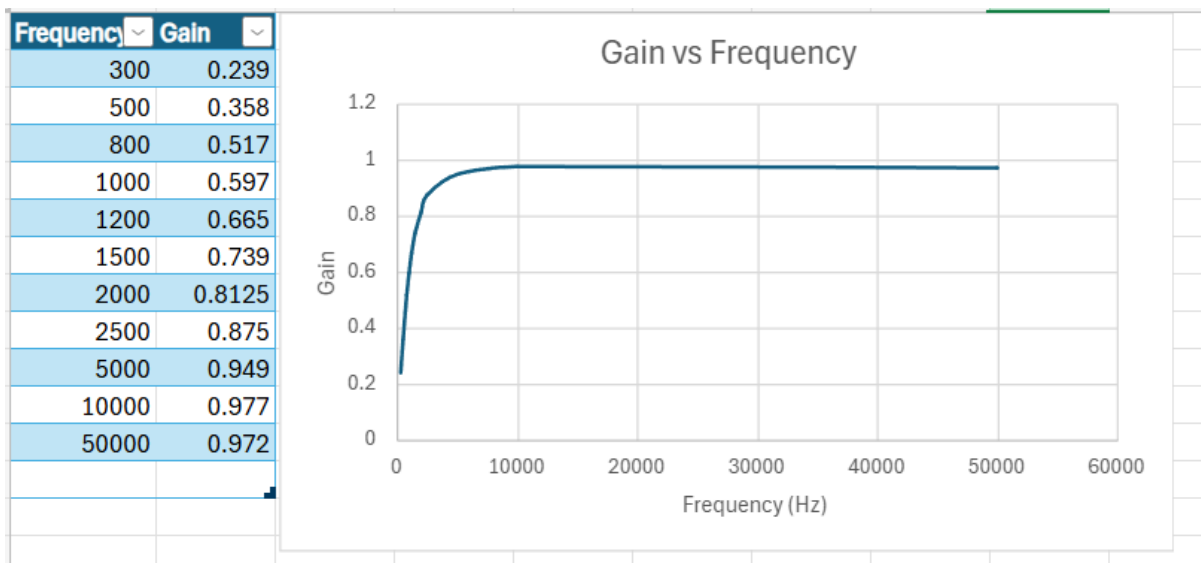
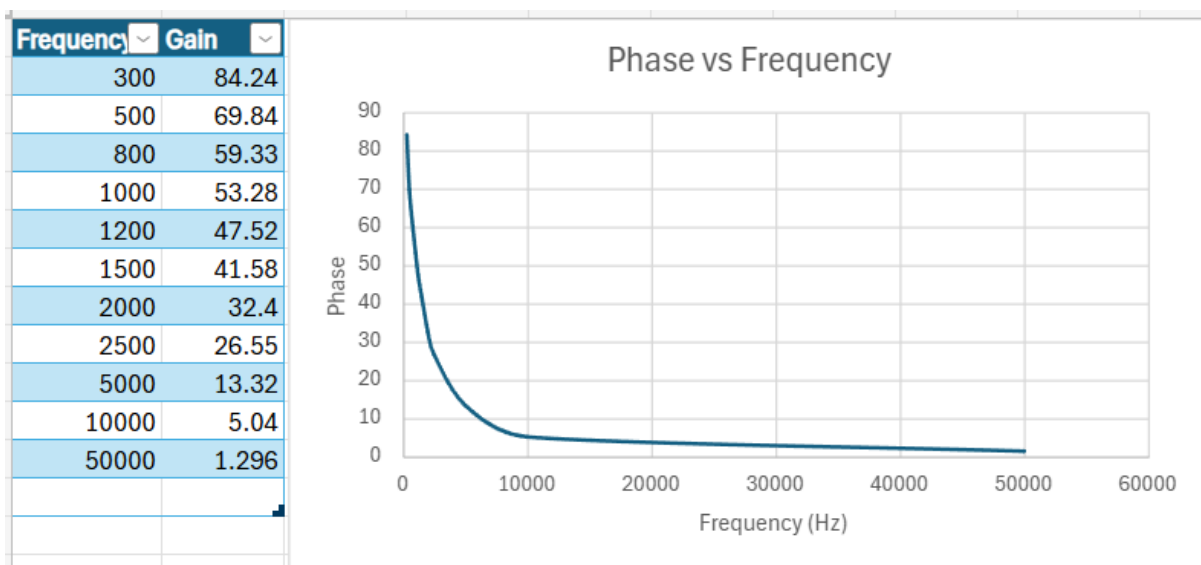


Figure 9.9: Phase Vs Frequency Graph for High Pass Filter

Gain vs Frequency:



Phase vs Frequency:



9.5. Band Pass Filter

By connecting or “cascading” together a single Low Pass Filter circuit with a High Pass Filter circuit, we can produce another type of passive RC filter that passes a selected range or “band” of frequencies that can be either narrow or wide while attenuating all those outside of this range. This new type of passive filter arrangement produces a frequency selective filter known commonly as a Band Pass Filter.

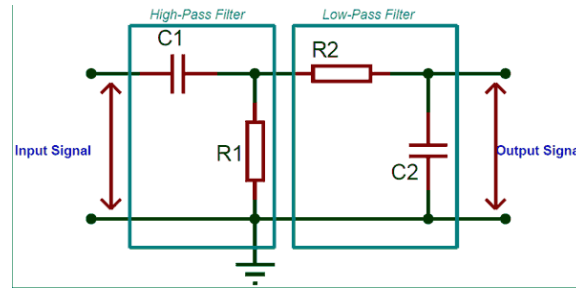


Figure 9.10: Circuit Schematic of Band Pass Filter

The first half of the circuit is a High-Pass filter which filters the low frequencies and allows only the frequency that is higher than the set high cut-off frequency. The value of this high cut-off frequency can be calculated using the formulae:

$$f_H = \frac{1}{2\pi R_1 C_1}$$

The second half of the circuit is the Low-Pass filter circuit which filters the higher frequencies and allows only the frequency that is lower than the set low cut-off frequency. The value of low cut-off frequency can be calculated using the formulae:

$$f_L = \frac{1}{2\pi R_2 C_2}$$

This type of filter circuit is called as 2nd order filter because it has two resistors and two capacitors. A band pass filter could be a 2nd order filter or of higher order since a minimum of two resistor and capacitor is needed for proper functioning of the circuit. Now, when an input signal frequency is supplied to this filter it outputs a frequency which is higher than $f_{c(low)}$ and lower than $f_{c(high)}$. In other words, the output frequency can be given by $f_H - f_L$, the frequency that lies in between this region is called as bandwidth. Hence the Bandwidth of the filter can be calculated by:

$$Bandwidth = f_L - f_H$$

9.5.2 Frequency Response of Band Pass Filter

The graph is plotted against the input frequency in the X-axis and the output in decibels in the Y-axis. When the input frequency is less than the lower cut-off frequency (f_L) the output remains less than -3dB and when it exceeds that frequency, the output reaches the maximum and stays there until the frequency exceeds the higher cut-off frequency (f_H). The peak at which the output gain stays maximum is called as the resonant frequency. It is simply the geometric mean of the upper higher cut-off frequency and the lower cut-off frequency. The formulae to calculate the same is given below:

$$Center\ Frequency\ (f_{center}) = \sqrt{f_L * f_H}$$

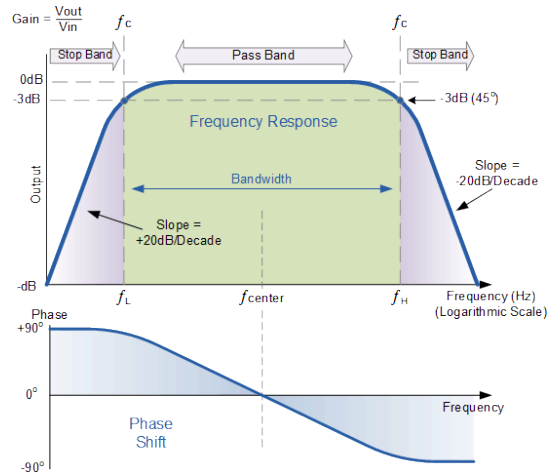


Figure 9.11: Frequency Response of Band Pass Filter

Task 3: To construct passive Band Pass Filter and verify with appropriate simulation.

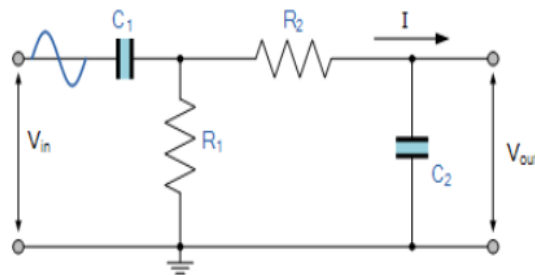


Figure 9.12: Circuit Diagram of Passive Band Pass Filter

- Design a band pass filter that will only allow a range of frequencies to pass above 1kHz (1,000Hz) and below 30kHz (30,000Hz). Assuming that both the resistors have values of 10k Ω , calculate the values of the two capacitors required. Provide the calculations in box below:

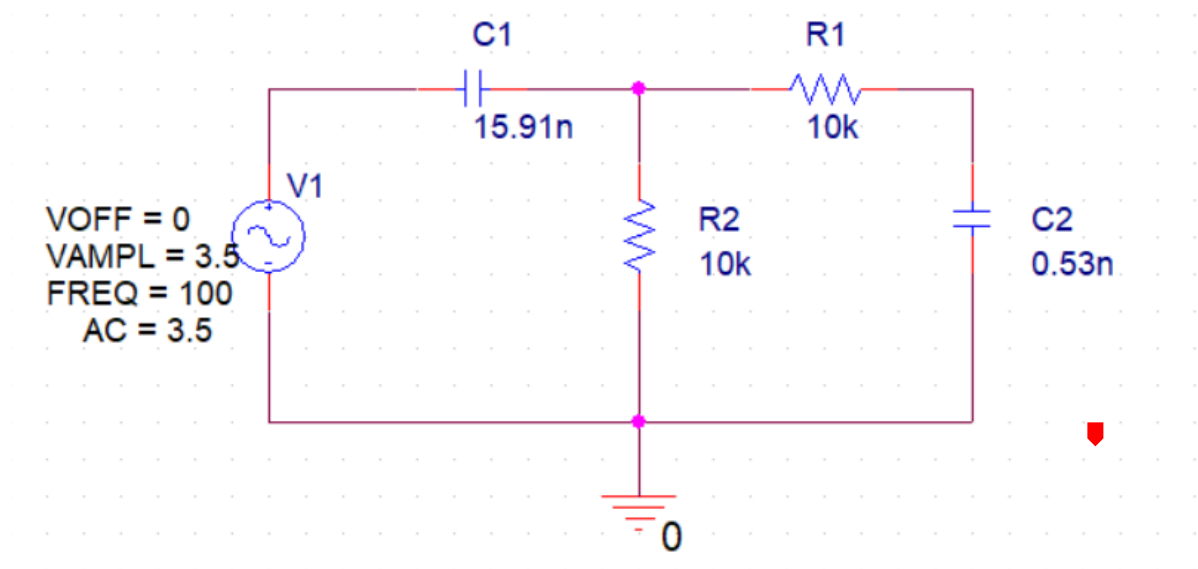
$C1 = 15.91\text{nF}$ using fH and $C2 = 0.53\text{nF}$ using fC.

2. Calculate the center frequency and the bandwidth of this band pass filter

$$f_c = \underline{\hspace{2cm}} 5476.11 \text{ Hz}$$

$$BW = \underline{\hspace{2cm}} 29000$$

3. Simulate the circuit in Figure 9.12 on PSpice using above calculated values.



4. Test the circuit by giving Input signal of $V_{in}=3.5\text{V}$ for the frequencies mentioned in table 9.3. Observe and record the output voltages across C_2 on below mentioned frequencies. You can use

any method of simulation to perform this task, doing AC sweep¹ saves time. Do not forget to attached snippet of your circuit and the plot.

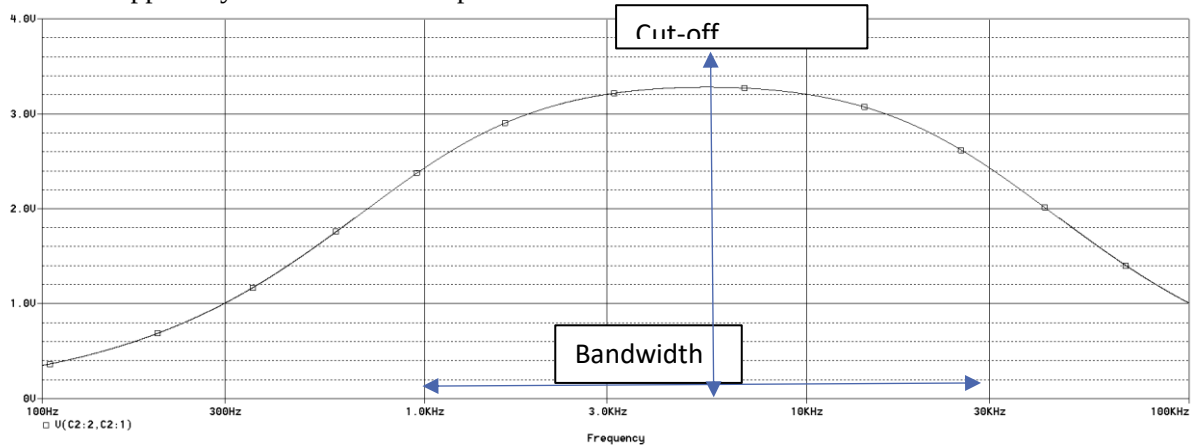


Table 9.3: Data Table for Band Pass Filter

Frequency (Hz)	V_{out}
100	348.022m
1k	2.4322
5k	3.2797
15k	3.0403
30k	2.4327
75k	1.2958
100k	1.0037

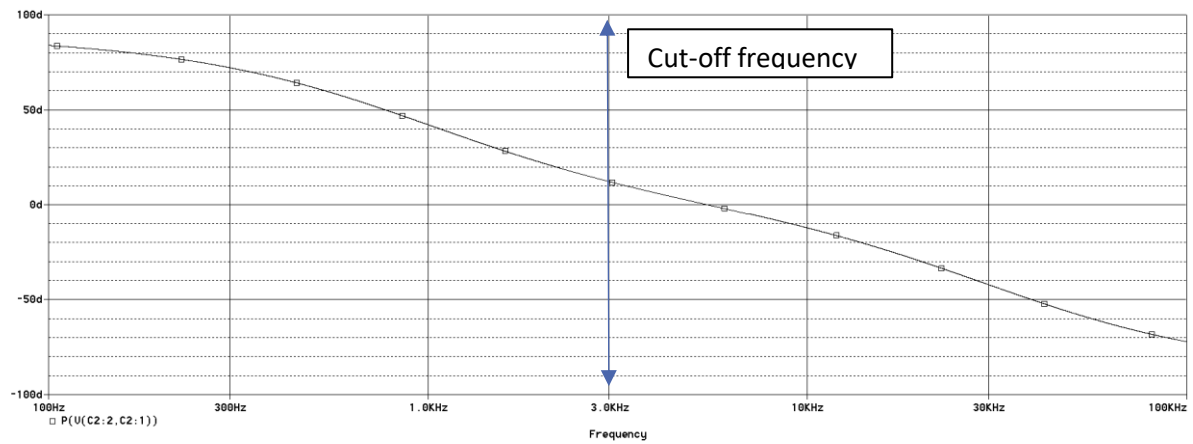
5. Also provide your analysis on these values.

Initially the filter does not allow the signal to pass however, as the frequency increases the signal is passed until it reaches its peak at 15kHz and after that the filter again stops the signal indicating it as a Bandpass filter allowing only certain range of signal to pass through.

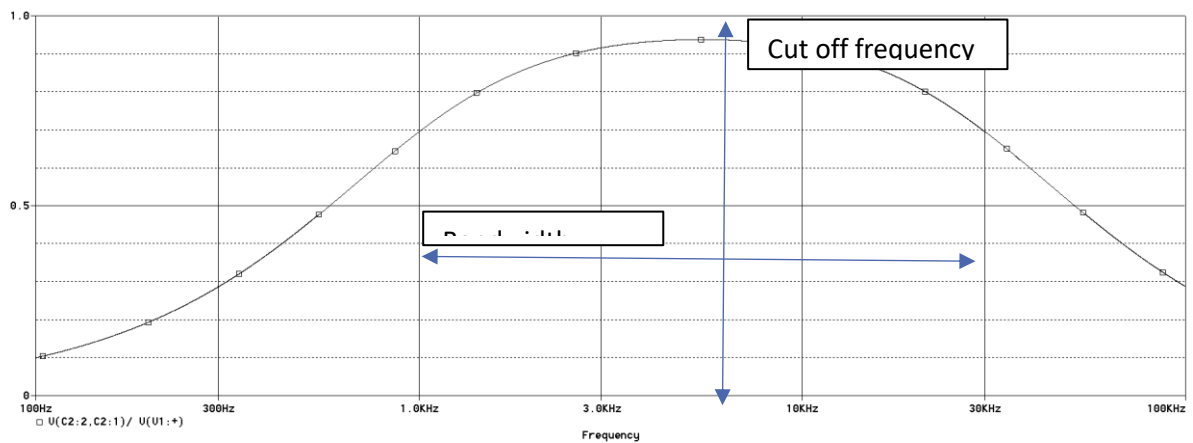
6. Plot the phase response and magnitude response of the filter and **label the cutoff frequency and highlight bandwidth on the plot**. Attach both plots.

¹ Perform AC sweep analysis with logarithmic scale for a suitable frequency range measuring output voltage across C_2

Phase Response:



Magnitude Response:





Assessment Rubric
Lab 9
Passive Filters

Name:	Student ID:
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Points Distribution

Task No.	LR 2 Simulation	LR 4 Data Collection	LR5 Results/Plots	LR 6 Calculation	LR10 Analysis	LR11 Design
Task 1	-	/15	-	/7.5		
Task 2	-	/15	-	/7.5		
Task 3	/5	/10	/15	/5	/5	/5
Total	/5	/40	/15	/20	/5	/5
CLO Mapped	CLO 2	CLO 2	CLO 2	CLO 2	CLO 2	CLO 2

Affective Domain Rubric		Points	CLO Mapped
AR 4	Report Submission	/10	CLO 4

CLO	Total Points	Points Obtained
2	90	
4	10	
Total	100	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.



Lab Evaluation Assessment Rubric

#	Assessment Elements	Level 1: Unsatisfactory Points 0-1	Level 2: Developing Points 2	Level 3: Good Points 3	Level 4: Exemplary Points 4
LR1	Circuit Layout	Connections between circuit components are mostly wrong. Circuit layout is cluttered. Needs guidance to make correct equipment/ component connections.	Few of the connections made between circuit components are wrong. Circuit layout is not neat and clean as per standards.	Circuit layout and connections are correct but not all connections are neat and clean as per standards.	Neat, clean and correct connections of all the circuit components/ equipment are made as per standard circuit diagram.
LR2	Program/Code/ Simulation Model/ Network Model	Program/code/simulation model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR3	Troubleshooting	Unable to identify the fault/minimal effort show in troubleshooting.	Able to identify the fault but unable to remove it.	Able to identify the fault but partially removes it.	Able to identify the fault and takes necessary steps and actions to correct it.
LR4	Data Collection	Measurements are incomplete, inaccurate and imprecise. Observations are incomplete or not included. Symbols, units and significant figures are not included.	Measurements are somewhat inaccurate and imprecise. Observations are incomplete or vague. Major errors are there in using symbols, units and significant digits.	Measurements are mostly accurate. Observations are generally complete. Minor errors are present in using symbols, units and significant digits.	Measurements are both accurate and precise. Data collection is systematic. Observations are very thorough and include appropriate symbols, units and significant digits and task completed in due time.
LR5	Results & Plots	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR6	Calculations	Formulae used and/or calculations are incorrect. Units are not mentioned with results.	Formulae used are correct but there are few mistakes in calculation which lead to incorrect results.	Formulae used and end results are correct. Complete working steps are not shown. Proper units are not mentioned with results.	Formulae used, end results and all calculations are correct with all the intermediate steps clearly shown. Units are mentioned with results.
LR10	Analysis	Results are not interpreted or are analyzed incorrectly.	Analysis of the obtained results is incomplete. Conclusions are not drawn.	Results are analyzed and concluded but are not well explained and justification is not provided with the help of reasoning.	Results are well analyzed supported with reasons. Good comparison is made between the theoretical and experimental results with correct and useful conclusion.
LR11	Design	Proposed design is unsubstantiated or does not satisfy most of given constraints. The breakdown of system into features is incomplete and still at lower resolution.	Justification for proposed design is weak, and it only satisfies some constraints. The breakdown of system is missing some features and resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies most given constraints. The breakdown of system into features seems complete, but resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies all given constraints. The breakdown of system into features seems complete and at appropriate resolution, given students' level.
AR4	Report Submission	Not accepted after 1 week.	Late submission after 2 days and within 1 week.	Late submission after the lab timing and within 2 days of the due date.	Timely submission of the report and in the lab time.